

EX RICO 23 WHITE BEAN

Ex Rico 23 is a white (navy) bean (*Phaseolus vulgaris* L.) adapted to field bean production in southwestern Ontario. It exhibits much more tolerance to white mold (*Sclerotinia sclerotiorum* (Lib.) de Bary) than other cultivars, along with good agronomic performance and cooking quality.

Ex Rico 23 est un petit haricot blanc (*Phaseolus vulgaris* L.) adapté à la production du haricot sec dans le sud-ouest de l'Ontario. Il possède beaucoup plus de tolérance à la moisissure blanche (*Sclerotinia sclerotiorum* (Lib.) de Bary) que les autres cultivars actuellement en usage, ainsi que de bonnes qualités agronomiques et culinaires.

Phaseolus vulgaris L. 'Ex Rico 23' was obtained from Centro Internacional de Agricultura Tropical (CIAT) in 1976, as part of the first International Yield and Adaptation Nursery. It has been designated CIAT line 'P-756', although it is a public cultivar in Latin America. The pedigree of Ex Rico 23 is unknown. A licence, No. 2055, was issued for Ex Rico 23 in December 1980.

Agronomic Performance

Ex Rico 23 was originally tested at the Elora Research Station in 1976, where it outyielded all other white bean cultivars. It has been tested in Ontario Field Bean Variety Trials since 1977. Data on agronomic performance and cooking quality are shown in Table 1. Over the 4-yr period 1977-1980, it has outyielded Fleetwood, Kentwood and Seafarer by 6, 11, and 24%, respectively.

Ex Rico is between Fleetwood and Kentwood in maturity and is adapted to areas with approximately 2900 corn heat units, which represents the warmer part of the main white bean-growing area in southern Ontario. Its seed size and shape are acceptable to the white bean industry. The major advantage of Ex Rico 23 over other licenced cultivars is good field tolerance to white mold caused by *Sclerotinia sclerotiorum* (Lib.) de Bary. Values for white mold infection in Table 1 indicate reaction to the disease in three trials where white mold was severe. In these three trials, Ex Rico 23 outyielded the other three cultivars by from 27 to 169%. In trials with-

out white mold infection, Ex Rico 23 did not outyield Fleetwood. Details of the tolerance of Ex Rico 23 to white mold and the effect on its yield relative to other cultivars are presented by Tu and Beversdorf (1981).

Description

Ex Rico 23 is a bush type, somewhat more spreading than Kentwood but less spreading than Fleetwood. It tends to produce short vines in some years. It has white flowers, light buff pods, and white, glossy seed coats. A larger proportion of the flowers than in other cultivars are produced within the leaf canopy and fewer flowers are above the leaf canopy. Ex Rico 23 has a larger tap root than other cultivars, as well as larger nodules and higher N₂ fixation rates under comparable conditions.

Quality

Taste panel evaluations conducted by the Agriculture Canada Research Station, Harrow (Table 1) and by Libby, McNeill, and Libby, Ltd., Chatham, Ontario, rated Ex Rico 23 as acceptable and comparable with the check cultivars Sanilac, Seafarer and Kentwood. Protein content was also similar to the check cultivars.

Disease Reaction

Ex Rico 23 is resistant to races 1 and 15 of bean common mosaic virus, and tolerant to the alpha, beta and gamma races of anthracnose (*Colletotrichum lindemuthianum* Sacc. and Magn.), but susceptible to the delta race. It is distinctly more field-tolerant to white

Table 1. Comparative data for Ex Rico 23, Fleetwood, Kentwood and Seafarer (Ontario Field Bean Variety Trials, 1977-1980)

Cultivar	Yield (t/ha) (21) [‡]	Maturity (days) (19)	Seed weight (g/100 seeds) (14)	Cooking quality score [‡] (4)	White mold (% of plants infected) (3)
Ex Rico 23	2.95	99	19.8	9.1	15.4
Fleetwood	2.77	101	18.9	10.2	40.3
Kentwood	2.66	92	22.4	9.5	63.3
Seafarer	2.38	88	19.0	9.0	60.6

[‡]No. of tests.[‡]Values shown are the totals for appearance, flavor, and texture of canned beans, with each characteristic rated from 1 (poor) to 5 (excellent).

mold than other cultivars grown in Ontario (Table 1).

Seed Distribution

Breeder seed will be maintained by the Crop Science Department, University of Guelph, Guelph, Ontario, and released through the Ontario Seed Stock Distribution Committee of the Canadian Seed Growers' Association.

TU, J.C. and BEVERSDORF, W.D. 1981. Tolerance to white mold [*Sclerotinia sclerotiorum*

(Lib.) de Bary] in Ex Rico 23, a cultivar of white bean (*Phaseolus vulgaris* L.). Can. J. Plant Sci. **62**: (in press).

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